Lab Documentation: Build a Book Inventory App

Project Overview

Built a responsive Book Inventory web application using HTML5 and CSS3 as part of the freeCodeCamp Responsive Web Design Certification. The app structures tabular data and dynamically applies custom background styling, status badges, and star-rating indicators using advanced CSS attribute selectors and structural pseudo-classes.

Implementation Workflow

1. HTML5 Structure & Data Formatting

- Built a semantic HTML table using `<table>`, `<thead>`, `<tbody>`, `<tr>`, `<th>`, and `<td>` elements to display book metadata (Title, Author, Category, Status, Rate).

- Assigned status classes (`read`, `to-read`, `in-progress`) to table rows (`<tr class="...">`) to drive row-level conditional styling.

- Embedded status badges using `<span class="status">` inside table cells.

- Represented ratings using a container `<span class="rate [one|two|three]">` housing nested `<span>` elements for individual indicator dots.

2. Advanced CSS Styling & Selector Engineering

- **Row-Level Gradients**: Applied horizontal linear gradients to table rows based on exact class matches using attribute selectors:
  - `tr[class="read"]`: `#d4edda` to `#c3e6cb`
  - `tr[class="to-read"]`: `#f8d7da` to `#f5c6cb`
  - `tr[class="in-progress"]`: `#fff3cd` to `#ffeeba`

- **Contextual Badges**: Styled `.status` spans using descending hierarchical attribute selectors (e.g., `tr[class="read"] span[class="status"]`) paired with vertical linear gradients (`to bottom`) and solid borders.

- **Dynamic Rating System**:
  - Styled rating circles using circular borders (`border-radius: 50%`), explicit dimensions (`15px` x `15px`), and neutral background fallback (`#eee`).
  - Used CSS Attribute Matchers and Pseudo-Classes to fill rating indicators dynamically:
    - `span[class~="one"] > span:first-child`: Fills the first dot.
    - `span[class~="two"] > span:nth-child(-n+2)`: Fills the first two dots using formulaic counting.

- span[class~="three"] span: Fills all three dots.

Knowledge Gained & Key Technical Takeaways

- **CSS Attribute Selectors**:
  - [class="value"]: Matches exact class string.
  - [class^="value"]: Matches classes starting with specified text (e.g., span[class^="rate"]).
  - [class~="value"]: Matches a specific word within a whitespace-separated class list (e.g., span[class~="two"]).
- **Functional Structural Pseudo-Classes**:
  - Mastered span:nth-child(-n+2) to target elements from index 1 up to n, allowing clean batch selections without extra HTML markup.
- **Visual Hierarchy & Depth**:
  - Combined directional CSS gradients (to right for rows, to bottom for badges/ratings) to build visual depth across tabular data.

CODE:















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Book Inventory

Title	Author	Category	Status	Rate
The Three Musketeers	Alexandre Dumas	Historical Novel	Read	
The Plague	Albert Camus	Philosophical Novel	To Read	
The Metamorphosis	Franz Kafka	Novella	Read	
Dead Souls	Nikolai Gogol	Picaresque	Read	
Lord of the Flies	William Golding	Allegorical Novel	In Progress	
Do Androids Dream of Electric Sheep?	Philip K. Dick	Science Fiction	Read	